

Why We Use the Sample Mean: A Minimum-Variance Derivation

Handout SOK-3081

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1. Motivation

Every introductory statistics course tells you that if you want to estimate an unknown population mean μ from a random sample, you should use the **sample mean** $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. But *why* the simple average, and not some other weighted combination of the observations?

This handout answers that question properly: we show that among all **linear, unbiased** estimators of μ , the sample mean is the one with the **smallest variance**. This is not a coincidence — it is a special case of the Gauss–Markov theorem you will meet again when we get to regression (where it justifies OLS as BLUE: Best Linear Unbiased Estimator).

Learning objectives. By the end of this handout you should be able to:

1. Set up a general *class* of estimators rather than committing to one formula in advance.
2. Impose unbiasedness as a constraint on that class.
3. Use constrained optimization (Lagrange multipliers) to find the variance-minimizing member of the class.
4. Connect the result to the Gauss–Markov theorem.
5. Verify the result numerically in R.

2. Setup

Let $X_1, X_2, \dots, X_n$ be a random sample — that is, the X_i are **independent and identically distributed (iid)** — with

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2, \quad i = 1, \dots, n.$$

Both μ and σ^2 are unknown population parameters. Our goal is to estimate μ using a function of the data.

Key idea: instead of guessing that “the average” is the right formula, we let the data speak. We consider the whole *family* of linear estimators

$$\hat{\mu} = \sum_{i=1}^n w_i X_i,$$

where $w_1, \dots, w_n$ are constants (weights) that we are free to choose. The simple average is just the special case $w_i = 1/n$ for all i — but nothing yet forces us to pick that. We will *derive* it instead.

3. Step 1: Impose unbiasedness

An estimator is **unbiased** if its expected value equals the true parameter, $E[\hat{\mu}] = \mu$, for *any* value of μ . Let’s see what this requires of the weights.

Using linearity of expectation,

$$E[\hat{\mu}] = E\left[\sum_{i=1}^n w_i X_i\right] = \sum_{i=1}^n w_i E[X_i] = \sum_{i=1}^n w_i \mu = \mu \sum_{i=1}^n w_i.$$

For this to equal μ for every possible value of μ , we need

$$\boxed{\sum_{i=1}^n w_i = 1.}$$

This is our **constraint**. Notice it does *not* pin down the individual weights — there are infinitely many weight vectors that sum to one (e.g. $w_1 = 1, w_2 = \dots = w_n = 0$ is also unbiased — that estimator just throws away all information except X_1 !). Unbiasedness alone is not enough; we need a second criterion to choose *among* the unbiased estimators.

4. Step 2: Compute the variance of an arbitrary linear estimator

Because the X_i are independent, the variance of a sum is the sum of variances (cross terms vanish):

$$\text{Var}(\hat{\mu}) = \text{Var}\left(\sum_{i=1}^n w_i X_i\right) = \sum_{i=1}^n w_i^2 \text{Var}(X_i) = \sigma^2 \sum_{i=1}^n w_i^2.$$

So the variance of any linear unbiased estimator is proportional to $\sum_i w_i^2$. **Smaller $\sum w_i^2$ means a more precise (lower-variance) estimator.** This gives us a clean optimization problem.

5. Step 3: Minimize variance subject to unbiasedness

We want to solve:

$$\min_{w_1, \dots, w_n} \sum_{i=1}^n w_i^2 \quad \text{subject to} \quad \sum_{i=1}^n w_i = 1.$$

This is a constrained optimization problem, so we use a **Lagrange multiplier** λ and form the Lagrangian

$$\mathcal{L}(w_1, \dots, w_n, \lambda) = \sum_{i=1}^n w_i^2 - \lambda \left(\sum_{i=1}^n w_i - 1 \right).$$

First-order conditions. Differentiate with respect to each w_i and set to zero:

$$\frac{\partial \mathcal{L}}{\partial w_i} = 2w_i - \lambda = 0 \quad \implies \quad w_i = \frac{\lambda}{2}, \quad \text{for every } i = 1, \dots, n.$$

This is the crucial step: the first-order condition is *identical for every i* , so **every optimal weight must be the same constant**, $w_i = \lambda/2$. The optimization itself — not an assumption — tells us the weights must be equal.

Apply the constraint. Substitute $w_i = \lambda/2$ into $\sum_i w_i = 1$:

$$\sum_{i=1}^n \frac{\lambda}{2} = n \cdot \frac{\lambda}{2} = 1 \quad \implies \quad \lambda = \frac{2}{n}.$$

Therefore

$$w_i^* = \frac{1}{n} \quad \text{for every } i = 1, \dots, n.$$

Second-order condition (confirming a minimum). The Hessian of the objective $\sum w_i^2$ is $2I_n$ (twice the identity matrix), which is positive definite. The objective is strictly convex, so the stationary point we found is indeed the unique **global minimum**, not a maximum or saddle point.

6. Step 4: Identify the estimator

Plugging $w_i^* = 1/n$ back into $\hat{\mu} = \sum_i w_i X_i$:

$$\hat{\mu}^* = \sum_{i=1}^n \frac{1}{n} X_i = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

The sample mean is the minimum-variance linear unbiased estimator of μ . Its variance is

$$\text{Var}(\bar{X}) = \sigma^2 \sum_{i=1}^n \left(\frac{1}{n}\right)^2 = \sigma^2 \cdot n \cdot \frac{1}{n^2} = \frac{\sigma^2}{n}.$$

This is the familiar formula for the variance of the sample mean — but now you have seen *why* no other linear combination of the data can beat it.

7. A quick alternative proof (Cauchy–Schwarz)

For intuition, here is a one-line alternative that avoids calculus. By the Cauchy–Schwarz inequality,

$$1 = \left(\sum_{i=1}^n w_i \cdot 1 \right)^2 \leq \left(\sum_{i=1}^n w_i^2 \right) \left(\sum_{i=1}^n 1^2 \right) = n \sum_{i=1}^n w_i^2,$$

which rearranges to

$$\sum_{i=1}^n w_i^2 \geq \frac{1}{n},$$

with **equality if and only if** all the w_i are equal (the equality condition of Cauchy–Schwarz), i.e. $w_i = 1/n$. This confirms both the bound $\text{Var}(\hat{\mu}) \geq \sigma^2/n$ and that the sample mean is the unique minimizer.

8. A second route to the same answer: least squares

The derivation above answers the question “which *linear combination* of the data has the smallest variance, given that it must be unbiased?” There is a completely different — and in some ways more elementary — way to arrive at exactly the same formula. Instead of starting from a class of estimators, we start from a **criterion of fit**.

Setup. Treat μ as a free parameter (not yet assumed to be the population mean) and define the **sum of squared deviations**

$$S(\mu) = \sum_{i=1}^n (y_i - \mu)^2.$$

This measures how far the data $y_1, \dots, y_n$ are, in total, from a single candidate “center” μ . The least squares principle says: choose μ to make this total squared distance as small as possible.

Minimize. Differentiate with respect to μ and set the derivative to zero:

$$\frac{dS}{d\mu} = \frac{d}{d\mu} \sum_{i=1}^n (y_i - \mu)^2 = -2 \sum_{i=1}^n (y_i - \mu) = 0.$$

Divide by -2 and expand the sum:

$$\sum_{i=1}^n (y_i - \mu) = 0 \quad \implies \quad \sum_{i=1}^n y_i - n\mu = 0.$$

Solve for μ :

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}.$$

Second-order condition. $\frac{d^2 S}{d\mu^2} = 2n > 0$ for any $n \geq 1$, so $S(\mu)$ is a strictly convex (upward-opening) parabola in μ , and the stationary point we found is indeed the unique minimum — not a maximum or an inflection point.

Two roads, one destination. Notice what just happened: minimizing *variance* over the class of linear unbiased estimators (Sections 3–6) and minimizing the *sum of squared deviations from a candidate center* (this section) are conceptually different problems — one is about the sampling behavior of an estimator across hypothetical repeated samples, the other is purely about fit to the one sample you have in front of you — and yet they deliver the identical formula, $\bar{y}$. This is not a coincidence, and seeing *why* it isn't is one of the more satisfying moments in this course.

Why this matters more than it looks. The least-squares derivation you just did is not a curiosity — it is a one-variable rehearsal for OLS regression. There, instead of fitting a single constant μ to the data, you will fit a line $\hat{y}_i = \beta_0 + \beta_1 x_i$, and minimize

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

using exactly the same trick: differentiate, set to zero, solve. In fact, if you set $\beta_1 = 0$ (no explanatory variable, just a constant), the regression problem collapses precisely to $S(\mu)$ above, and OLS “regression on a constant only” returns $\hat{\beta}_0 = \bar{y}$. So this handout has secretly already shown you the $\beta_1 = 0$ special case of OLS.

9. Why this matters for the course

This little derivation is a miniature version of the **Gauss–Markov theorem**, which is proved in matrix form in the textbook. There, $\hat{\mu} = \bar{X}$ is replaced by $\hat{\beta} = (X'X)^{-1}X'y$, “linear unbiased” becomes “linear unbiased in the regression coefficients,” and the conclusion becomes “OLS is BLUE” under the classical assumptions (exogeneity, homoskedasticity, no autocorrelation). The logic — restrict attention to a class of estimators, impose unbiasedness, minimize variance — is exactly the same.

10. Checking it numerically in R

Rather than trusting the algebra blindly, let's confirm the result by brute force: draw many random weight vectors that satisfy the unbiasedness constraint $\sum w_i = 1$, compute the theoretical variance $\sigma^2 \sum w_i^2$ for each, and check that none of them beats $w_i = 1/n$.

```
set.seed(1)

n      <- 10          # sample size
sigma2 <- 4           # true variance of each X_i (arbitrary for illustration)

# Function to compute Var(mu_hat) = sigma^2 * sum(w_i^2) for a weight vector
var_linear_estimator <- function(w, sigma2) {
  sigma2 * sum(w^2)
}

# Weights for the sample mean
w_mean <- rep(1/n, n)
var_mean <- var_linear_estimator(w_mean, sigma2)

# Generate many random *unbiased* weight vectors: draw n-1 free weights,
# set the last one so that the weights sum to 1
n_trials <- 20000
random_variances <- numeric(n_trials)

for (t in 1:n_trials) {
  w_free  <- runif(n - 1, min = -1, max = 1.5)  # arbitrary random weights
  w_last  <- 1 - sum(w_free)                    # enforce sum(w) = 1
  w       <- c(w_free, w_last)
  random_variances[t] <- var_linear_estimator(w, sigma2)
}

cat("Variance of the sample mean:      ", round(var_mean, 4), "\n")
cat("Smallest variance found by search:", round(min(random_variances), 4), "\n")
cat("Any estimator beat the mean?      ", any(random_variances < var_mean - 1e-8), "\n")
```

Running this, you should see that `var_mean` equals $\sigma^2/n = 4/10 = 0.4$, and that **no randomly generated unbiased linear estimator ever beats it** — confirming the theoretical result above.

We can also visualize the trade-off directly in the simple $n = 2$ case, where the constraint $w_1 + w_2 = 1$ means we can plot variance as a function of a single free weight w_1 (with $w_2 = 1 - w_1$):

```

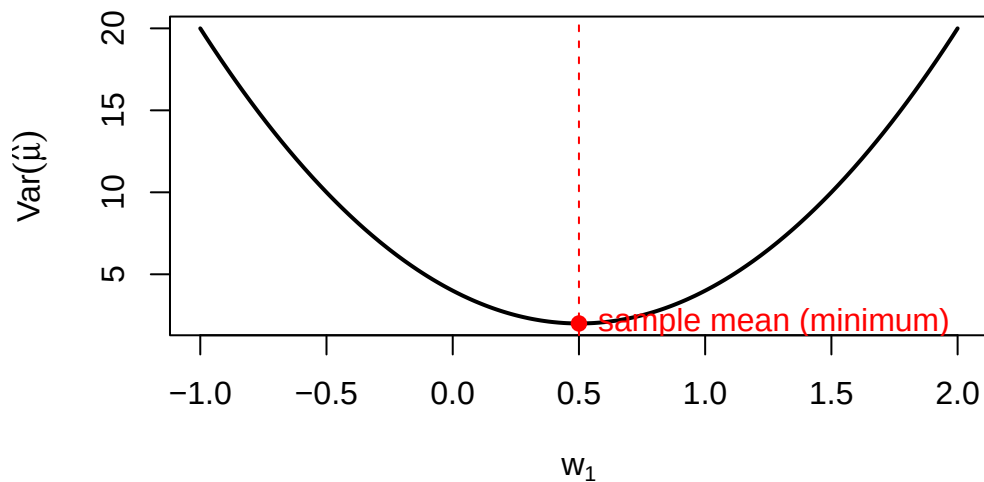
n2      <- 2
sigma2  <- 4

w1_grid <- seq(-1, 2, by = 0.01)
w2_grid <- 1 - w1_grid
var_grid <- sigma2 * (w1_grid^2 + w2_grid^2)

plot(w1_grid, var_grid, type = "l", lwd = 2,
     xlab = expression(w[1]), ylab = expression(Var(hat(mu))),
     main = "Variance of a linear unbiased estimator, n = 2")
abline(v = 0.5, col = "red", lty = 2)
points(0.5, sigma2 * (0.5^2 + 0.5^2), col = "red", pch = 19)
text(0.5, sigma2 * (0.5^2 + 0.5^2), labels = "sample mean (minimum)",
     pos = 4, col = "red")

```

Variance of a linear unbiased estimator, $n = 2$



The plot is a parabola with its unique minimum exactly at $w_1 = w_2 = 0.5$ — the sample mean — visually confirming the Lagrange multiplier result.

We can run the same kind of check on the least-squares criterion $S(\mu)$ from Section 8 — sweep over candidate values of μ , compute $S(\mu)$ for each, and confirm the minimum sits exactly at $\bar{y}$:


```

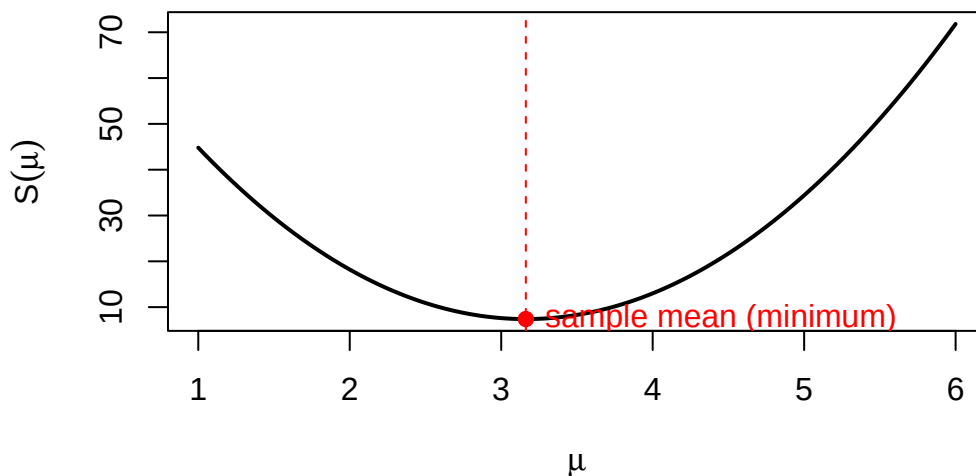
y <- c(2.1, 3.4, 1.9, 5.0, 2.7, 3.3, 4.1, 2.8) # example data
n <- length(y)

mu_grid <- seq(1, 6, by = 0.01)
S <- sapply(mu_grid, function(mu) sum((y - mu)^2))

plot(mu_grid, S, type = "l", lwd = 2,
      xlab = expression(mu), ylab = expression(S(mu)),
      main = "Sum of squared deviations vs. candidate mu")
abline(v = mean(y), col = "red", lty = 2)
points(mean(y), sum((y - mean(y))^2), col = "red", pch = 19)
text(mean(y), sum((y - mean(y))^2), labels = "sample mean (minimum)",
      pos = 4, col = "red")

```

Sum of squared deviations vs. candidate mu



```
cat("Value of mu that minimizes S(mu) by grid search:", mu_grid[which.min(S)], "\n")
```

Value of mu that minimizes $S(\mu)$ by grid search: 3.16

```
cat("Sample mean, mean(y):", round(mean(y), 4), "\n")
```

Sample mean, $\text{mean}(y)$: 3.1625

You should see the grid-search minimizer match `mean(y)` almost exactly (up to the resolution of the grid), and the plot should again be a clean convex parabola — same shape as the w_1 plot above, but now the horizontal axis is the candidate value of μ itself rather than a weight.

11. Summary

Step	What we did
1	Defined the class of linear estimators $\hat{\mu} = \sum w_i X_i$
2	Imposed unbiasedness $\Rightarrow \sum w_i = 1$
3	Wrote variance as $\sigma^2 \sum w_i^2$ (using independence)
4	Minimized $\sum w_i^2$ subject to $\sum w_i = 1$ via Lagrange multipliers
5	Found $w_i^* = 1/n$ for all i , i.e. $\hat{\mu}^* = \bar{X}$
6	Confirmed with Cauchy–Schwarz and numerically in R
7	Defined $S(\mu) = \sum (y_i - \mu)^2$ and minimized via $dS/d\mu = 0$
8	Got the <i>same</i> answer, $\hat{\mu} = \bar{y}$, from a completely different criterion (least squares, not variance)

Takeaway: the sample mean is optimal in (at least) two independent senses. It is the **minimum-variance** linear unbiased estimator (Sections 3–7) — a statement about how the estimator behaves across hypothetical repeated samples — and it is the **least-squares** estimator (Section 8) — a statement about fit to the one sample you actually have. These are different questions that happen to share an answer. The least-squares route is the one to keep especially close at hand: it is the exact one-variable rehearsal for how you will derive the OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ later in the course, right down to the “differentiate and set to zero” mechanics.